Bashir Agro Admin Panel
Phase 8: Notification System Module

Phase 8 Scope

Implement ONLY Notification System after Phase 7.

Included:
- Database notifications
- Admin notification center
- Read/unread management
- Notification UI
- Event-based notification structure
- Notification preferences preparation
- Tests

Not included:
- Flutter push notification
- Firebase Cloud Messaging
- SMS gateway
- Email campaigns
- Mobile app notifications

Objective

Create a centralized notification system inside the Laravel Admin Panel.

The purpose is to inform Admin/Owner about important system events.

Examples:

Farmer created
Location created
Task assigned
Task completed
Assignment status changed
Important system actions

Notifications must be database-driven.

Notification Architecture

System Event

↓

Laravel Event

↓

Notification Class

↓

Database Notification

↓

Admin Notification Center

↓

User Reads Notification

Database Design

Use Laravel database notifications.

Create:

notifications

Fields:

id
type
notifiable_type
notifiable_id
data
read_at
created_at
updated_at

The notification data will store:

title
message
module
action URL
additional information

Notification Model

Use Laravel default:

Illuminate\Notifications\DatabaseNotification

Configure User model:

use Notifiable;

User can receive notifications through database channel.

Notification Events

Create notifications for:

1. Farmer Events

farmer.created
farmer.updated
farmer.deactivated

2. Location Events

location.created
location.updated

3. Task Template Events

task.created
task.updated

4. Assignment Events

assignment.created
assignment.updated
assignment.completed
assignment.cancelled

Notification Classes

Create:

app/Notifications/

Examples:

FarmerCreatedNotification.php

TaskAssignedNotification.php

AssignmentCompletedNotification.php

Each notification should contain:

title
message
module
link
timestamp

Trigger Workflow

Example:

Admin assigns task

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TaskAssignmentController store()

↓

Create Assignment

↓

Send Notification

↓

Save Notification Record

↓

Show in Admin Panel

Notification Center

Create:

resources/views/notifications/index.blade.php

Display:

- Title
- Message
- Module
- Date
- Read/Unread status
- Action button

Notification Dropdown

Update Admin topbar.

Show:

Unread notification count

Latest notifications

Mark as read action

Replace static notification badge values.

Routes

Create:

GET /admin/notifications

POST /admin/notifications/{id}/read

POST /admin/notifications/read-all

Protect routes with authentication middleware.

Controller

Create:

NotificationController.php

Methods:

index()

markAsRead()

markAllAsRead()

destroy()

Keep controller simple.

Unread Count

Create dynamic unread count:

User notifications where:

read_at IS NULL

Example:

```
Auth::user()
->unreadNotifications
->count();
```

Notification UI Requirements

Preserve Bashir Agro design:

- Bengali UI
- Green theme

- Existing topbar
- Cards
- Tables
- Responsive layout

Do not redesign dashboard layout.

Notification Preferences Preparation

Prepare structure for future settings:

Examples:

Enable/disable:

Task notifications

Assignment notifications

System notifications

Do not create complex settings panel yet.

Security

Use:

- Authentication middleware
- Authorization
- User ownership checks

A user should only read their own notifications.

Performance

Implement:

- Pagination
- Latest notification queries
- Database indexes if required

Avoid loading all notifications at once.

Testing

Create:

tests/Feature/NotificationTest.php

Test:

- Admin receives notification
- Notification appears in list
- Unread count works
- Mark as read works
- Mark all as read works
- Unauthorized users cannot access notifications
- Pagination works

Antigravity Prompt

```
Proceed to PHASE 8 ONLY: Notification System Module.

Use the existing Bashir Agro Laravel project after Phase 7.

Do not redesign the UI.

Do not implement:

- Flutter push notifications
- Firebase
- SMS
- Email campaigns
- Mobile notifications


Goal:

Create a database-driven Laravel notification system.


Implement:

- Laravel database notifications
- Notification classes
- Notification controller
- Notification center
- Read/unread system
- Topbar notification count
- Event-based notification triggers
- Feature tests


Preserve:

- Bengali UI
- Green agricultural theme
- Existing dashboard design


Database must be the source of truth.

Remove static notification counters.
```

```
After implementation:

Run:

php artisan test

Run:

vendor/bin/pint


Report:

- Files created
- Files modified
- Database changes
- Routes added
- Test results


STOP after Phase 8.

Do not start Phase 9 automatically.
```